

Algebra II with Trigonometry

Chapter 7.1

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4564659_alg-2trig-h-chp-7-1-note.pdf](#)

Sure! Here are 10 mental math questions based on the core trigonometric identities and concepts from the PDF:

1. Reciprocal Recall What is the reciprocal of $\sin(x)$?

2. Quotient Quick Check What is $\tan(x)$ in terms of $\sin(x)$ and $\cos(x)$?

3. Pythagorean Power If $\sin^2(x) = 1/4$, what is $\cos^2(x)$?

4. Even/Odd Identity Test What is $\sin(-x)$ in terms of $\sin(x)$?

5. Cofunction Swap What is $\sin(\pi/2 - x)$ equal to?

6. Product Property If $\tan(x) \cdot \cot(x) = ?$, what is the value?

7. Identity Application If $\tan^2(x) + 1 = y$, what is y in terms of $\sec(x)$?

8. Express in Sines & Cosines Write $\sec(x)$ in terms of $\cos(x)$.

9. Common Denominator Insight What is $1 / (1 + \tan^2(x))$ equal to in terms of $\cos(x)$?

10. Expression Simplification If $\sin(x) = 0$, what is $\cos^2(x)$?

Let me know if you'd like answer keys or explanations!